

White Paper: IP Convergence Based On SIP - Enhanced Person-To-Person Communications

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SIP

NOKIA

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1 Introduction

Nokia envisions a mobile life where mobility has pervaded all facets of everyday life. In this world users will have the ability to access, create, consume, and share digital content using interoperable devices that are connected with each other anytime, anywhere. IP convergence plays a major role in this vision.

IP convergence is the fusion of mobile and Internet domains, using packet-based networks as a common infrastructure, to enrich the way we communicate. With IP protocols it is possible to enrich the end-user experience by combining voice, video, text, and content seamlessly in person-to-person communication. This experience is further enhanced by bridging the existing gap between mobile and Internet domains and enabling access to services independent of the type of access (e.g., General Packet Radio Service [GPRS], LAN, WLAN).

Connectivity is becoming more widespread and diverse. Wireless access technologies such as Wideband CDMA (WCDMA), CDMA, GSM Enhanced Data Rates for Global Evolution (EDGE), and WLAN are emerging in a broad range of products, and the bandwidth provided by these new access types is continuously increasing. At the same time, the use of broadband connections in homes is growing rapidly. All of these access technologies will have one thing in common — IP. With IP as a common platform, the various access technologies can be seamlessly combined to create a completely new user experience.

A key element in IP convergence is the end-user experience: the value and potential of combining availability with rich conversation and sharing of content to provide enriched services. Even in the world of IP convergence, end users will still perceive person-to-person communications as the most important service. From the end user's perspective, the primary benefit of IP in communications is integration and interaction of services along with network-independent addressing. Voice, video, text, content sharing, and presence are all components of personal interactive communications, which can interact with each other to create services that are both beneficial and easy to use. Therefore, the ability to use one common protocol for all services will offer significant benefits to end users.

Brand new applications are already beginning to emerge now that the convergence of IP networks is starting to happen. For example:

- Push-to-talk over Cellular (PoC) is a new voice application enabled by SIP and 2G GPRS networks.
- Real-time video sharing is an emerging application for sharing live video or video clips in real time during a normal voice call also enabled by SIP, but in 3G networks.

Other interesting applications will emerge as developers around the world are provided with the tools to develop SIP-based applications to run in converged networks.

To maximize the true potential of IP convergence, the highly innovative developer community must be actively involved. Its indisputable creativity must be harnessed by being given access to the tools, in the form of open APIs and SDKs, to create SIP-based applications. Applications can be quickly developed due to the reuse of common IP protocols, and quickly deployed by utilizing the common connectivity features of SIP to establish connections.

In the following chapters, key issues in IP convergence will be highlighted. The concept of IP convergence and the reasons driving it will be discussed. Also, the concepts of real-time video sharing, PoC, and other services, as different components of personal communications and the technology behind these services will be covered.

2 Why IP Convergence?

2.1 Person-to-Person Communications

In the world of IP convergence, services and applications will be redefined. Applications will no longer be isolated entities exchanging information only with the user interface. The next generation of applications will involve peer-to-peer connectivity between IP-capable terminals. The ability to establish a peer-to-peer connection between new IP-enabled mobile devices is the key ingredient for enabling richer person-to-person communication. Once an IP-level connection is established between users, there are a whole range of IP protocols that can be used to exchange information and experiences between end users. PoC is a good example of what can be achieved with IP-based applications.

In order to communicate, IP-based applications must have a mechanism to reach the correspondent. Today, fixed and mobile telephone networks perform this critical task of establishing a connection. By dialing the other user's telephone number, the network can establish an ad hoc connection between any two terminals. This critical connectivity capability still does not exist widely in the Internet.

SIP-based session management complemented by critical mobile network capabilities (i.e., authentication, roaming, and network interconnectivity provided by the 3GPP IP Multimedia Subsystem [IMS]) is the service infrastructure required. With such a system in place, it is possible to establish an IP connection between two terminals. Once the connection is established, the IP connection can be used to exchange all types of communication media (voice, video, content, etc.).

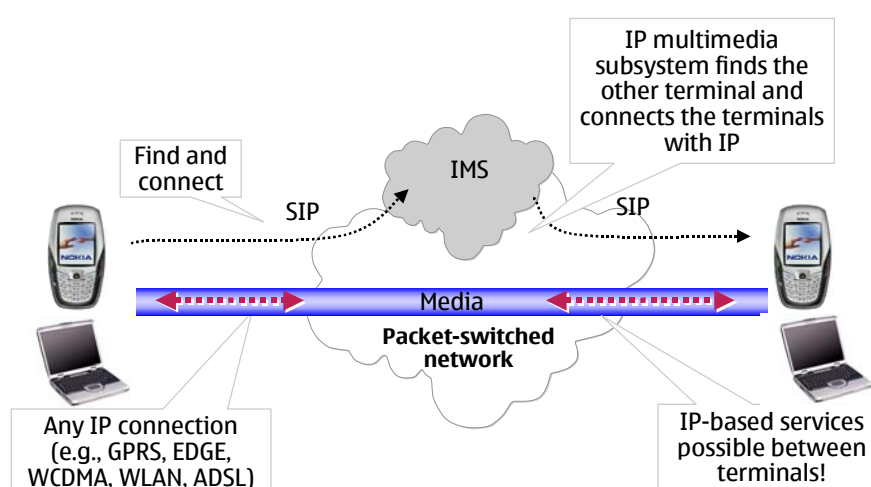


Figure 1: Person-to-person connectivity in IP convergence networks

2.2 Fast, Easy Service Creation

The phenomenal growth of the Internet and the Web has put the spotlight on IP-based networks and the inherent advantages of IP technology.

In IP convergence networks, the creation platforms are also based on IP protocols. All services, including communications services, are created with Internet-based protocols and languages. The benefit is that when all services are created with open and standard IP protocols, the creation and integration of these services is much faster and easier than those based on telecom protocols.

One of the key benefits of IP convergence is the possibility of giving developers the opportunity to develop new applications. SIP-enabled person-to-person applications can be built on open SIP APIs. This will allow new person-to-person applications to be developed and deployed with speeds not previously possible.

Operators are in the best position to offer added value to their subscribers. First, they have the subscriptions and profile data of the users. Second, they own the provisioning and billing machinery. Third, they control the end-to-end QoS for services. And fourth, they have access to the mobile-network-specific enablers such as location and presence information with the user's permission.

Service creation will occur in three identified domains:

- **Network operators.** Network operators can offer services that fully exploit the capabilities of their IP network and service capability servers integrated to the solution. The innovation of third-party developers can be used to quickly develop and deploy new, exciting services.
- **Third-party service providers.** Third-party service providers or corporations can use SIP capabilities by interfacing with the operator network server using an external application server. It is also possible for a third-party service provider to operate a SIP application server itself.
- **End users.** End users can create services by using the capabilities offered by an operator or a third-party service provider. For example, a Web interface capable of building up services using a simple Call Processing Language (CPL) could be offered to end users.

Since the underlying elements of SIP are so similar to HTTP, creating network-based services such as time-dependent call forwarding is quick and straightforward. Millions of software developers can design and implement new SIP-based voice services just as quickly and easily as they develop Web pages.

By not requiring major hardware upgrades to application servers, but rather enabling new software-based services using SIP, service providers can reduce the time associated with deploying new features from months to days. This means ever-improving communications services for subscribers.

2.3 Connectivity and Service Mobility Increase User Benefits

The number of access technologies is continuously growing. Differences such as wide and local area mobility vs. fixed-access methods will remain. Although from a broad perspective all networks converge into a single IP network, different networks inherently have some characteristics that differ. Securing the best quality interaction between networks is needed for seamless end-to-end services.

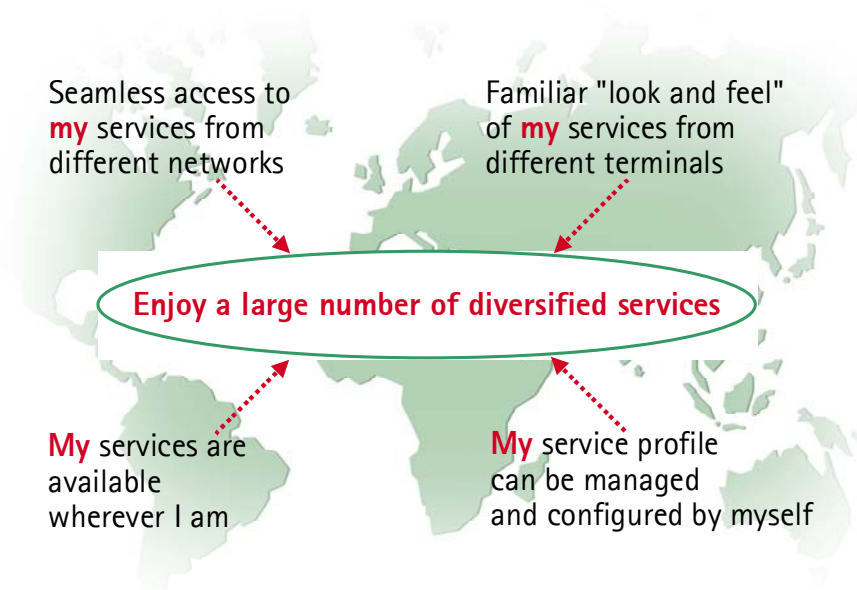


Figure 2: Service mobility creates a unified service experience

Service mobility means the accessibility of a person's personalized services and his/her personal reachability through any access network or terminal. User interfaces and services are independent of time and place, but are at the same time customized according to the terminal used and are aware of place and time. For example, a session (e.g., a videoconference) could be started on a PC, continued on a mobile terminal, and concluded on a television screen at home.

When all services (both browsing-type mobile Internet services and person-to-person communication services) are IP based, they can be offered through any access network – the operator can create services in one platform and offer them via multiple-access networks, thus greatly increasing the addressable customer base and serving the existing base better. Services that are accessible wherever, whenever, and through whichever device will increase customer loyalty and reduce churn. Service mobility and personal reachability is achieved with SIP and IPv6 Mobility. IPv6 gives a device a universal address, enabling it to be reached from any network.

End users benefit from SIP by having the same address while choosing the most suitable interface — mobile terminal, PC, PDA, etc. While SIP is the generic protocol over all access networks, end users may use the most capable access network available, for example in terms of price, bandwidth, usability, etc. Services are also personalized, which makes the user experience appealing.

2.4 New Business Opportunities

Applications in IP convergence can no longer be viewed as stand-alone. End users will undergo rich experiences due to several components interacting with each other. From the end user's perspective, the primary benefit of IP in communications is integration and interaction of services. Voice, video, instant messaging, and presence are all different aspects of personal interactive communications, each of which can also interact with the others to create services that are both beneficial and easy to use.

Historically, service integration has been difficult in mobile networks, since they all required their own protocols, standards, and service domains. In the IP convergence era, service integration is easy, since services are created within one domain. It is easy to integrate voice-based applications with content sharing applications, mobile terminal applications with PC applications, and so on. The possibilities are endless. In IP convergence networks, the operator needs only one network to take care of, while SIP remains the enabling protocol.

IP convergence provides opportunities for new revenue streams by enabling new services that were not previously possible or were too complex and expensive to implement. The IP convergence infrastructure enables new flexible billing models, that weren't possible before. The same charging model — price per minute and message — can be maintained for voice telephony and messaging, i.e., for non-real-time applications. Initially, there may be more package-based billing or service combinations. Package-based billing is the most suitable for real-time applications such as SIP-based instant messaging.

The management and operational costs of running a communications service are vastly reduced by using SIP, since new functionalities, such as real-time video sharing, presence, and instant messaging can be added easily to an existing SIP network, avoiding the need to run and build a completely separate, parallel network to support each feature. The equipment costs themselves are reduced, since only one set of proxy and location servers are needed, rather than two. System capacity is improved, because resources can be shared across many services. Infrastructure investments in firewalls, mobility services, etc., which have been put in place for supporting SIP, can also be reused.

3 New Paradigm Based on Previous Generation's Legacy

IP convergence builds on well-established service paradigms, for example Short Message Service (SMS) and Multimedia Messaging Service (MMS), by adding new functionality and new content and service types in user-understandable steps. Because end users can relate to the new services as enhanced service experiences from existing networks, the barriers for adopting them will be significantly reduced, leading to rapid adoption and high penetration.

The market will develop towards IP Convergence services as an evolution. SMS has been the main data service so far, but today MMS, browsing, downloading and other services are taking off rapidly. The service offering will further evolve in the IP convergence era with real-time services such as real-time video sharing and Push to talk over Cellular.

In the development of third-generation mobile networks and beyond, the focus has shifted towards the IP paradigm. In particular, the work so far has focused on the adoption of IP transport, IP mobility, and VoIP protocols (e.g. SIP) in mobile networks.

In the future, IP will allow the provisioning of multimedia services and of other services, such as instant messaging, Web browsing, etc., in an integrated fashion. In particular, users will no longer be forced to see the cellular network as capable of providing only voice services or Internet access. Current attempts to evolve the service infrastructure have simply tried to apply the current concepts and services to basic services, e.g., voice services, without considering the importance that service integration will assume.

An IP-based service architecture, capable of supporting multimedia and service integration, will be a key element of future cellular networks. Also, current service architecture models, as they are evolving at present, do not meet the requirements for service integration, from a technology point of view. Therefore, a distributed IP-based technology architecture approach is needed. In practice, this means the reuse of existing Internet Engineering Task Force (IETF) protocols, when possible, which will allow for an easier deployment and acceptance of the service architecture, both in the cellular and IETF environments.

4 Technology Behind IP Convergence: SIP

Session Initiation Protocol (SIP) will be the standard signalling protocol/mechanism to support multimedia sessions in 3GPP Release 5 networks and beyond. This choice ensures that future multimedia services will be truly Internet based. SIP is defined by the IETF Session Initiation Protocol working group in RFC3261 and is very different from traditional signalling methods used in telecommunication networks. It shares much with existing Internet protocols such as HTTP and SMTP (protocol header formats).

Like HTTP, SIP is a text-based client-to-server protocol. SIP was designed to establish, modify, and terminate multimedia sessions or calls. Clients send requests to servers and servers send responses back. A request invokes a method on the server. However, SIP is different from HTTP in the way that a SIP terminal (User Agent, UA) may act as both a client and a server. Therefore "client-to-client" communication is possible. The fact that it is text based makes SIP relatively easy to implement and debug. SIP is also an extensible and modular protocol. The modularity helps to implement SIP clients in different device categories, from mobile devices to personal computers.

SIP service architecture follows the Internet's successful design. Services are developed in the application layer and the network is used exclusively for data transport. SIP service architecture also takes into consideration the growing power of terminals by moving more responsibility to communication end points. The ultimate goal is to push service logic to the end points, leaving service providers and end users to develop and control the services. As a consequence, standard interfaces to develop services are needed and are provided by service development tools familiar from the Internet world, such as SIP/Common Gateway Interface (CGI; an API protocol) and SIP/Servlet (Java™ Servlet) servers on the application server side. Actual Web developers will be able to use their experience to develop and integrate new communication services

SIP also enables IN- and ISDN-type services as well as user mobility. And SIP supports instant conferences, adding parties to full mesh and enabling multicasting, and uses global Uniform Resource Identifier- (URI) based addressing that is similar to Internet e-mail (SIP:user@domain, where "user" is a user name or a telephone number and "domain" is a domain name or a numeric address).

In short, by implementing SIP, the operator gets the whole array of IP-based services in the same network, and they are easy to implement, manage, and control. New service development and introduction are made simple, compared to other protocols.

4.1 SIP and Open Standards

Open standards are essential to enable mass deployment of mobile multimedia services and to prevent market fragmentation, by maximizing the number of potential customers using compatible solutions. This applies to all players in the mobile service business — the content industry, application developers, operators, and manufacturers alike. Open standards cooperation is done in standardization bodies such as the 3rd Generation Partnership Project (3GPP), 3rd Generation Partnership Project 2 (3GPP2), Internet Engineering Task Force (IETF), and Open Mobile Alliance (OMA).

For example, 3GPP has defined the architecture of multimedia coding technologies and formats based on open standards. The overall structure is presented in Figure 4.

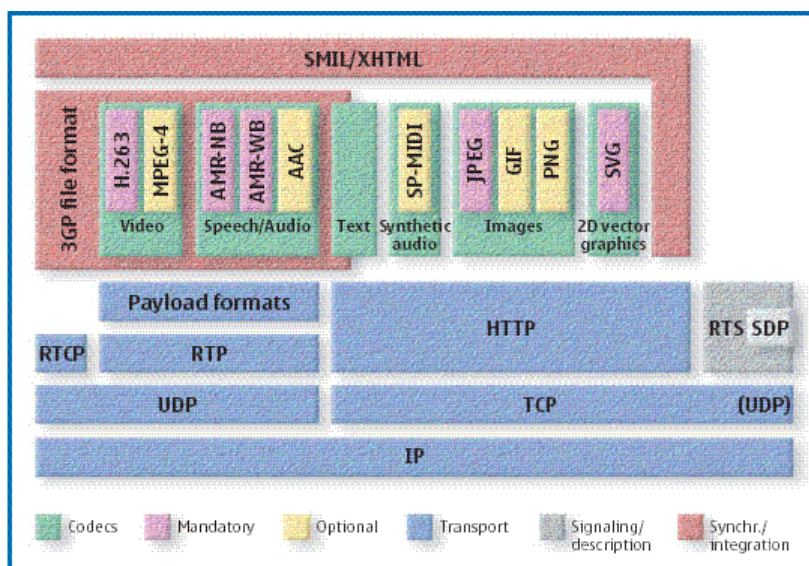


Figure 3: 3GPP multimedia content management

3GPP has adopted this architecture as part of the specification of two essential multimedia applications, the Multimedia Messaging Service (MMS) [REF: 3GPP TS 26.140] and the Packet Switched Streaming Service (PSS) [REF: 3GPP TS 26.234]. For instance, a video clip in a 3GPP media file can be composed from H.263 video and AMR-NB audio tracks.

When multimedia applications are implemented on embedded and constrained devices such as mobile devices, it is important to optimize the number of codecs and formats that need to be supported. This has been a leading design constraint for the 3GPP team, which made sure that the selection of codecs and file format for MMS and PSS is aligned. In this way, interoperability between MMS and PSS content and multimedia architecture, as well as future multimedia applications, will be ensured.

4.2 SIP and IPv6

The widely deployed version of the Internet Protocol, IPv4, was designed for a much smaller number of Internet hosts than the Internet is experiencing today. The IPv4 address space problem will become even greater when hundreds of millions (or billions) of cellular handsets are connected to the Internet. Insufficient IPv4 address space has been one of the main drivers to specify IP version 6 (IPv6). IPv6 is a mature technology that will be adopted in cellular networks and terminals, and is implemented by various manufacturers today. The most important benefit of IPv6 is the very large address space.

The IPv4 address space has been extended by the use of private IPv4 addresses together with Network Address Translation (NAT). Use of private IPv4 addresses and NATs is very common in GPRS networks today. However, NAT breaks the end-to-end nature of the Internet, and restricts the services used in the Internet. A public IP address is especially important for SIP-based peer-to-peer services, such as real-time video sharing and gaming. The general IPv4 NAT cannot be used for SIP-based services. This is due to the fact that in SIP, the signalling and the media stream follow a different route. The SIP signalling carries IP addresses for the inbound media streams within the Session Description Protocol (SDP). In the case of end points behind IPv4 NATs, the IP addresses that are exchanged within the SDP are the private IP addresses that are only valid within the address space of the NAT.

In summary, IPv6 provides a robust, future-proof platform for SIP-based services. SIP and IPv6 enable users to originate, receive, and access services from any end node and any location. From the end-user's point of view, IPv6 and SIP enable ubiquitous communication in person-to-person, as well as client/server environments. For services this means increased availability and usage, while more devices are capable of being online.

4.3 SIP and Quality of Service (QoS)

In short, Quality of Service (QoS) means providing a consistent, predictable data delivery service. In other words, satisfying customer application requirements. QoS refers to the ability of a communication system to provide an appropriate transport service for delivering relevant traffic stream types of different users. It can sometimes be difficult to define the exact technical parameters required to ensure such delivery, because the end user's perceptions of a "service quality" may differ from one user to another. QoS only manages bandwidth according to application demands and network management settings, and in that regard it cannot provide certainty if it involves sharing. Hence, QoS with a guaranteed service level requires resource allocation to individual data streams. Clearly, QoS is an end-to-end issue, as it affects the terminal as well the network solutions. Any QoS assurances are only as good as the weakest link in the "chain" between sender and receiver.

More often, applications are based on quality, reliability, and timeliness assurances. In particular, services that use voice and video streaming, e.g., VoIP and real-time video sharing, are quite demanding in this sense. Operators can differentiate themselves from their competitors by offering different kinds of service packages that have different kinds of QoS levels. The requirements for transport service vary according to application stream types and end users.

In the future, it will be possible to broaden the scope of these QoS packages into a more application-driven one. QoS may be a parameter of the application, with the actual service created around that application. Then, if an end user decides to buy that particular service, s/he can do so whether or not s/he belongs to a certain service-level package (provided his/her terminal and access mode support the required QoS).

In summary, QoS does not create additional bandwidth. Instead, it manages the existing bandwidth in such a way as to allow for efficient transportation of traffic through the network, allowing for predictable delays, delay variation (jitter), and packet loss rates.

4.4 SIP and Java Technology

Java technology has become widely available in mobile terminals and provides a standard platform for application development. Due to the availability of Java in a wide range of mobile terminals it is the most suitable technology platform to implement SIP-based applications for mid and low-end terminals.

Work has been completed to define the use of SIP for Java applications. The SIP API for the Java 2 Platform, Micro Edition (J2ME™) has been specified by the Java Community Process in JSR-180. Java developers now have a standard and open tool kit to develop SIP-based Java applications.

4.5 SIP and HTTP

SIP and HTTP stacks have a lot in common. Advanced SIP implementation usually provides a simple HTTP stack. One example of HTTP and SIP cooperation is that after some SIP signalling, a large file is fetched over HTTP using the HTTP GET method. Moreover, a SIP client (e.g., mobile phone) may host a simple HTTP server and offer files, etc., so that other parties can download information from the client (embedded Web server).

SIP works well in conjunction with the Web (MIME, URI, DNS). De facto, a lot of SIP services could be originated by Web pages (click to call, Web phone book, etc.). SIP services can also use the resident client Web/WAP browser to present information (call presentation service, messages, pictures, etc.). And SIP enables "multicast Web," for example sending Web pages on the top of SIP.

5 Applications in IP Convergence

IP will greatly enhance person-to-person communications by combining multiple media types and communication methods with presence and community information, not only person-to-person but also instant multipoint-to-multipoint, i.e. conferencing. With IP, adding "richer media" (video streams, live video, or application sharing), presence (availability, preferred media type), or community information is straightforward, as all the services use the same communication protocol (IP) and signalling (SIP).

The applications described below are the first applications emerging in IP Convergence. Whereas voice, video and messaging services will remain to be the basis for communication they will be used in a more flexible manner, complemented by presence, location or other context-related information. The ultimate target is described in the rich call concept in which the building blocks for on-line conversations will be integrated to make a seamless end-user experience.

5.1 Real-Time Video Sharing

The introduction of real-time person-to-person video sharing during a call is a key evolutionary step for mobile video services. Sharing live video or video clips in real time during a normal voice call allows users to enrich their communication with ease. Users can add and remove the video element, as needed, spontaneously.

Person-to-Person Video Services Evolution

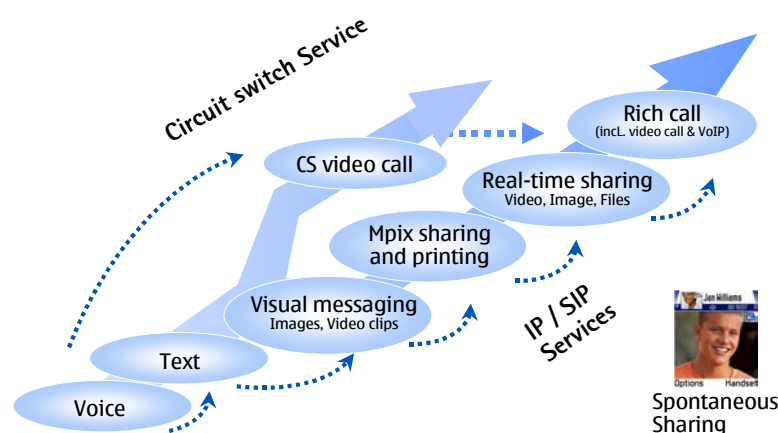


Figure 4: IP path enables evolution of person-to-person video communication towards rich call

Person-to-person video communication in mobile devices started with "video messaging" or video MMS, which is a natural evolution from still-image-based multimedia messaging. Video recording capability is available in a wider base of camera phones. The phone user can record a video clip and send it to another mobile device within a multimedia message. It can be opened and played back for viewing or saved to the device memory if desired and allowed.

"Real-time video sharing" enables mobile users to share one-way video in real time while they are already active in a voice session. This is a great way to gain additional revenue from voice calls and differentiate within 3G services. Wideband CDMA (WCDMA) operators can offer one-way real-time

video sharing as an enhancement to existing voice calls. Once Dual Transfer Mode (DTM) is available in EDGE networks and EV-DV in CDMA networks, they enable a similar use case for real-time video sharing as WCDMA networks.

“Video telephony” is about making or receiving a video call where the mobile user can see as well as talk to the other person. In effect, it allows the user visual and verbal communication. The conversation can be spiced up by seeing what the other person sees or by seeing the person on the other end. Video telephony will be supported in the first phase by circuit-switched technology, but the future IP path will evolve towards the rich call concept supporting IP-based video telephony.

Compelling Service

The evolution of packet-switched (PS) video services allows convergence with the PC domain, which will greatly increase the number of available end points. PS video services also have a clear evolution path for improved video quality, efficiency, and new service creation.

Real-time video sharing will produce a new stream of user-generated data traffic. This will increase the operator (packed data) revenues generated during voice calls.

Person-to-person sharing of real-time video during a voice call enables users to share the moment or their surroundings spontaneously. The following typical scenario is expected during 2005, with a WCDMA network. Lucy and Kate are close friends with active social lives. Lucy is on her way home from work. She calls Kate to see what her plans are for the evening. Kate tells Lucy that she is at a party. In order to enrich their voice call Kate starts to share her live camera view. As Lucy watches the live video stream, she can ask Kate to move the camera around so she can see all her friends having a great time. Kate ends the video sharing session, and they continue their conversation with the ongoing voice call.

Real-time video sharing is based on standardized 3GPP and IETF technologies. Use of standardized technologies will promote interoperability and speed up market take-off. Session management is accomplished utilizing SIP. Real-time video sharing uses common SIP infrastructure with other SIP-based applications. Employed codecs and other session-related parameters are negotiated between the end points using SIP/SDP during session establishment. In practice, this allows greater intelligence on the terminals, which can negotiate directly appropriate conditions for the session.

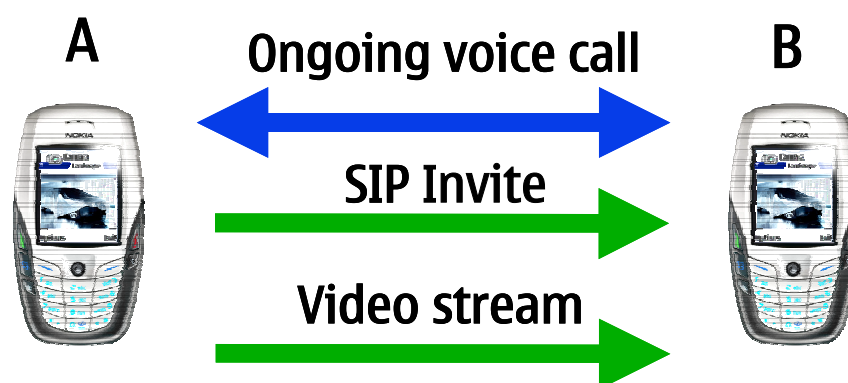


Figure 5: Real-time video sharing between two terminals

5.2 Push-to-Talk Over Cellular (PoC)

One of the first SIP-based applications emerging is Push-to-talk over Cellular (PoC). PoC makes two-way radio service possible over a cellular network, providing direct one-to-one and one-to-many voice communication service to cellular handsets.

The principle of the communication is simple: calls are started with the push of a key and connected directly — just push to talk. Push-to-talk calls are one-way communications — while one person speaks, the other(s) listens. With just one key press on the phone, the speakers take turns responding to each other on a first-come, first-served basis.

A Push-to-talk connection is typically connected virtually instantaneously. Thanks to the "always-on" connection, a single press reaches an active talk group. Users no longer need to make several calls to coordinate with a group. Additionally, PoC reserves cellular access and radio resources only one-way for the duration of talk spurts, instead of for the entire call session like in conventional telephony.

Nokia is actively participating in 3GPP, OMA, and other standardization forums, and promoting new services in mobile networks. The Push-to-talk service will be based on open standards to enable interoperability between different terminals and services. This offers terminal manufacturers an opportunity to implement the Push-to-talk facility across all GSM mobile phone categories.

Nokia has defined a PoC solution and adapted it to the GSM/GPRS environment. The solution will be open to other manufacturers, and Nokia's current plan is to adapt the solution in the next phase to the 3G environment. The Push-to-talk service will become a successful, open IP, standards-based service, complementing other IP multimedia communication services enabled by the 3GPP Release 5 IP Multimedia Subsystem (IMS).

5.3 Presence

As mobile communication becomes even more personal, presence will add a whole new dimension to staying in touch. Tomorrow's subscribers will know when and how they can reach their friends, even before they make a call.

With presence, mobile terminals will provide dynamic status information about subscribers, including their call/connection status, identities, terminal capabilities, location, and availability. The result will be a visible network personality, one that signals valuable information about the subscriber's reachability. In the fixed world, this information supports buddy lists and other similar services that let Internet users know when selected friends are on-line.

With presence, mobile subscribers will be able to show their status (busy, away from my desk, on holiday, etc.) to callers and suggest the most appropriate mode of communication (chat, instant message, e-mail, etc.).

Rich call concept is largely about combining availability, communication, and content, therefore presence information should also be seen as a user interface to rich-call services rather than just "instant messaging." Presence is an application in itself, but is also an enabler and enrichment to other applications such as rich call, messaging, and browsing. Also, presence enables call completion so that the caller can choose the best communication method with the callee, thus generating traffic in new ways that were hitherto not possible.

In its simplest form, a basic presence service might allow a user to publish a personal greeting that is visible to a caller prior to initiating the call.

Presence is a central constituent in the rich call concept. Presence information can be shown to rich-call users and thus influence their behavior (service and application selection), and also be used in the network to control the handling of rich calls based on pre-defined user preferences.

5.4 Messaging

Mobile messaging today predominantly means SMS, which has become a tremendous success over the past years, and voice messaging. MMS, an evolution from SMS, enriches messages with multimedia content, boosting mobile-to-mobile messaging to even higher levels. With MMS it is possible to

combine text with richer content types, photographs, images, audio and voice clips, and even video clips. MMS services have been launched by 230 operators around the world.

Instant messaging is an extremely popular application in the fixed Internet. Enhancing today's SMS, instant messaging enables messaging that is enhanced by presence information. Users are aware of each other's status, which provides better service and supports new types of use scenarios. Messages can be sent in real time to the receiving party whose on-line status can be checked with the presence feature. With SIP, the proven SMS business model may be implemented with new technology.

With SIP, instant messaging can be seamlessly integrated with conversational communications, thus creating the concept of ubiquitous communications — seamless usage of various real-time communications media and messaging services depending on the presence information of the parties.

Mobile instant messaging combines the success stories of SMS messaging (as being always reachable) and Internet-based instant messaging (which uses the presence information of the parties). SIP-based instant messaging paves the way for Integrated Messaging solutions, allowing users to combine the benefits of e-mail, MMS, and instant messaging, and choose the appropriate communication method according to the presence information of the receiving party.

5.5 Rich Call

Rich call is a concept for session-based communication, in the form of voice, video, and text, supported with parallel sharing of content (image, data, or other value-added information) between conversation participants. The rich call concept is a framework that will combine a common set of enablers and technologies, presence, PoC, real-time video sharing, messaging, etc., into a seamless and coherent user experience.

By joining the fixed and mobile worlds, SIP will enable true service mobility and access independence. It will also support the rich call concept, or concurrent access to images, data, or other value-added information during a single session. This will add a "see what I mean" capability for users through a combination of voice, real-time video sharing, messaging, content sharing, and other services. If SIP is the engine for the rich call concept, then common group management and presence are the catalysts, enabling subscribers to initiate the services and select from a range of different communication modes.

The key to the rich call concept is the end-user experience — the value and potential of combining availability with rich conversation and the sharing of content to provide enriched services. Users will be able to seamlessly switch between interaction modes, which will provide a more coherent user experience.

SIP provides the essential elements for a rich call before, during, and after the call. With SIP it is easy to implement, for example, presence-enhanced contacts and click-to-talk features, which enable easy connection establishment. During the call establishment itself, SIP enables sending a rich clip, which can be a picture with a call subject or a special ringing tone. During the call it is easy to share documents and generally implement information push features. Furthermore, during the session tear-down, the same kinds of call termination messages are possible as in call establishment. And finally, after the call it is still possible to send an instant message containing, for example, an advertisement.

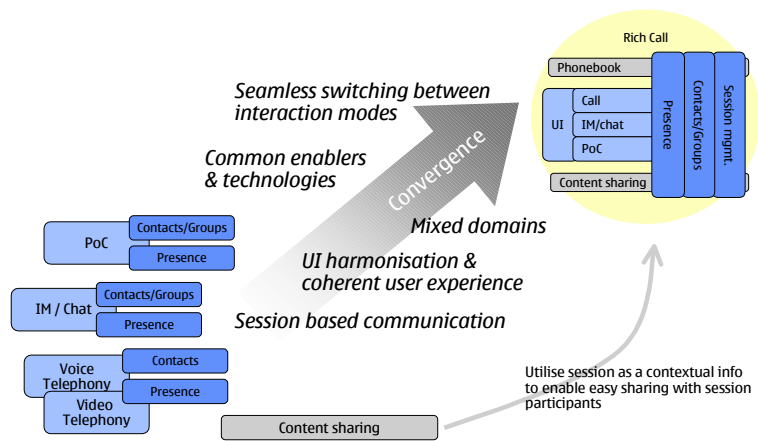


Figure 6: Rich-call concept

6 IP Convergence Is Happening Today

IP convergence is already happening. Today, GPRS connections provide connections to the Internet, and connectivity is rapidly expanding with the growth of WCDMA, CDMA, WLAN, ADSL, and other access technologies. IP is now widely accepted as the technology that will provide the platform for future services. And the first IP convergence applications (Real-Time Video Sharing [RTVS] and PoC) are already emerging in the market and will change the way we communicate.

IP convergence will play a significant role in making the Nokia vision come true. Nokia envisions a mobile life where mobility has pervaded all facets of everyday life. In this world users will have the ability to access, create, consume, and share digital content using interoperable devices that are connected with each other anytime, anywhere. This creates an increasing demand for bandwidth, connectivity features, and economy that cannot be served by circuit-switched mobile networks in the present form. Implementing this vision requires the kind of knowledge that Nokia has of the mobility business — both infrastructure and terminals — and the Internet.

Finally, to summarize, the key changes and benefits from the materialization of IP convergence are highlighted below:

- IP convergence will greatly enhance person-to-person communications by combining voice, video, and other interactive communications services, enriching the way we interact and communicate.
- The key to the rich call concept is the end-user experience — end users will benefit from the combination of availability, rich communication, and shared content to provide enriched services.
- New packet-based communication services such as Push-to-talk over Cellular and real-time video sharing are already emerging, and will enrich the way we communicate.
- Innovative new services will emerge as the use of Internet protocols is combined with peer-to-peer connectivity provided by SIP. This will be driven by the availability of SIP APIs and SDKs for the developer community.

7 Glossary

Term or abbreviation	Meaning
3GPP	3rd Generation Partnership Project
ADSL	Asymmetric Digital Subscriber Line
API	Application Programming Interface
CGI	Common Gateway Interface
CPL	Call Processing Language
DTM	Dual Transfer Mode
GIF	Graphics Interchange Format
GPRS	General Packet Radio System
HLR	Home Location Register
HTTP	Hypertext Transfer Protocol
IETF	Internet Engineering Task Force
IMS	IP Multimedia Subsystem
IN	Intelligent Network
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
ISDN	Integrated Services Digital Network
ISP	Internet Service Provider
JPEG	Joint Photographic Experts Group
LAN	Local Area Network
MMS	Multimedia Messaging Service
NAT	Network Address Translation
OMA	Open Mobile Alliance
PC	Personal Computer
PDA	Personal Digital Assistant
PoC	Push-to-talk over Cellular
QoS	Quality of Service
RTVS	Real-Time Video Sharing
SDK	Software Development Kit
SDP	Session Description Protocol
SIM	Subscriber Identity Module
SIP	Session Initiation Protocol
SMS	Short Message Service

SMTP	Simple Mail Transfer Protocol
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
XML	Extensive Markup Language
VoIP	Voice over IP
WCDMA	Wideband CDMA
WLAN	Wireless Local Area Network